Intersection of lines and planes

Basic skills

Q1. a) $\underline{r} = \begin{pmatrix} 5+\lambda \\ -2+2\lambda \\ 4-2\lambda \end{pmatrix}$ so $x = 5+\lambda$
$y = 2\lambda - 2$
$z = 4 - 2\lambda$

b) $\underline{r} = \begin{pmatrix} 7-2\mu \\ \mu \\ -3 \end{pmatrix}$ $x = 7-2\mu$
$y = \mu$
$z = -3$

c) $\underline{r} = \begin{pmatrix} 5+\lambda-5\mu \\ -2+2\lambda-2\mu \\ 4-2\lambda-\mu \end{pmatrix}$ $x = 5+\lambda-5\mu$
$y = -2+2\lambda-2\mu$
$z = 4-2\lambda-\mu$

Q2. Normal $\underline{n} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}$

so $-2x - y = d$

sub point $(5, 2, 1)$

$\Rightarrow -10 - 2 = d$

$d = -12$

so $-2x - y = -12 \Leftrightarrow 2x + y = 12$

Q3. $2+3\lambda = -4+6\mu$

$3+2\lambda = -10\mu$

$5+5\lambda = 8-3\mu$

$x(\lambda) = x(\mu)$

$y(\lambda) = y(\mu)$

$z(\lambda) = z(\mu)$

$$\Rightarrow \quad \mu = \frac{-3}{10} - \frac{2}{10}\lambda$$

so $$2+3\lambda = -4 - \frac{18}{10} - \frac{12}{10}\lambda$$

$$\Rightarrow \frac{42}{10}\lambda = -\frac{39}{5} \qquad \lambda = \frac{-13}{7}$$

$$\Rightarrow \mu = -\frac{3}{10} - \frac{2}{10} \times \frac{-13}{7} = \frac{1}{14}$$

Now check

$$5+5\times\frac{-13}{7} = \frac{-30}{7}, \qquad 8-3\times\frac{1}{14} = \frac{109}{14}$$

not consisdent so no Point of intersection

The lines L_1 and L_2 are skew

Competency skills

Q4. $\underline{r} = \begin{pmatrix} 2+\lambda \\ 3 \\ 5+5\lambda \end{pmatrix}$, $2x - y + z = 13$

a) $x = 2+\lambda$

$y = 3$

$z = 5+5\lambda$

b) $2(2+\lambda) - 3 + 5 + 5\lambda = 13$

$\Rightarrow \quad 7\lambda = 7 \quad \Rightarrow \quad \lambda = 1$

So $\overset{2+\lambda}{x = 3}$, $\overset{3}{y = 3}$, $\overset{5+5\lambda}{z = 10}$

$(3, 3, 10)$ is point of intersection

Q5.

$x = 4 + 4\lambda$

$y = -1 + 2\lambda \quad \Rightarrow \quad 3x + 2y + z = 16$ becomes

$z = 3 - \lambda$

$12 + 12\lambda - 2 + 4\lambda + 3 - \lambda = 16$

$\Rightarrow \quad 15\lambda = 3 \quad \lambda = \frac{1}{5}$

So $x = 4 + \frac{4}{5} = \frac{24}{5}$, $y = -1 + \frac{2}{5} = \frac{-3}{5}$ $z = 3 - \frac{1}{5} = \frac{14}{5}$

so point of intersection is

$$\left(\frac{24}{5}, \frac{-3}{5}, \frac{14}{5}\right)$$

Q6.

$x = 5+2t$

$y = 1$ so $y + 6z = 10 - 3x$

$z = -1-t$

$\Rightarrow 1 - 6 - 6t = 10 - 15 - 6t$

$\Rightarrow -5 - 6t = -5 - 6t \Rightarrow 0 = 0$

so true for all t. Hence all points on the line are in the plane

Q7.

$x = 2\lambda$

$y = 3-\lambda$ so $\underline{r} \cdot \underline{n} = 0$

$z = 4+3\lambda$

$$\Rightarrow \begin{pmatrix} 2\lambda \\ 3-\lambda \\ 4+3\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = 0$$

$\Rightarrow 4\lambda + 15 - 5\lambda - 4 - 3\lambda = 0 \Rightarrow 4\lambda = 11$

so $\lambda = \frac{11}{4} \Rightarrow x = \frac{11}{2}, y = \frac{1}{4}, z = \frac{49}{4}$

Advanced skills

Q8. L_1 is $\underline{r} = \begin{pmatrix} 2 \\ 3 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 1 \\ 2 \end{pmatrix}$

or $\begin{pmatrix} -4 \\ 0.5 \\ 1 \end{pmatrix}$

and so

$x = 2 - 8\lambda$

$y = 3 + \lambda$ so

$z = -5 + 2\lambda$

$3x + y - z = -5$

$\Rightarrow 6 - 18\lambda + 3 + \lambda + 5 - 2\lambda = -5$

$\Rightarrow -19\lambda = -19$

$\Rightarrow \lambda = 1$ and so $x = -6, y = 4, z = -3$

$(-6, 4, -3)$

Q9. planes are

$$\begin{array}{r} 5x + 2y = 0 \\ - \quad x + 2y - z = 0 \\ \hline 4x + z = 0 \end{array}$$

$\Rightarrow z = -4x$ and $x = -\frac{2}{5}y$

so $z = \frac{8}{5}y \Rightarrow -4x = \frac{8}{5}y = z$

is the line of points

Q10.

$$x = 1 + 2kt$$
$$y = 3 + t$$
$$z = 3 + 5t$$

$\Rightarrow$ $1 + 2kt + 3 + t - 3 - 5t = 1$

$\Rightarrow$ $1 + (2k - 4)t = 1$

To be contained all t must work so $2k - 4 = 0$ $\Rightarrow$ $k = 2$